R210517

Rajiv Gandhi University of Knowledge Technologies- RK Valley -AP

Academic Year 2024-25: Semester II - Mid Semester Test-III

Civil Engineering - 23CE2201- Hydraulics Engineering

R21 Batch& Regular Mode of Examinations

Year: ENGG-2 | **Date of Examination: 09-APRIL, 2025**

Time: 60 Minutes | **Max Marks: 15**

3X5 Marks= 15

The Question Paper Consists of 2 units. Answer one full question from each unit. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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1. The internal and external diameter of an impeller of a centrifugal pump which is running at 1450 rpm are 150mm and 350mm respectively. The discharge through pump is 0.08 m3/sec and the velocity of flow is constant and equal to 0.8 m/s. The diameters of the suction and delivery pipes are 80mm and 60mm respectively and suction and delivery heads are 6m (abs.) and 30m (abs.) of water respectively. If the outlet vane angle is 45^0 and power required to drive the pump is 56.186 kW, determine i) Vane angle of the impeller at inlet. ii) the overall efficiency of the pump, and iii) manometric efficiency of the pump.

OR

2. Show that for a given discharge, specific energy is minimum when $Q^2.T / gA^3$ is unity, where T is the water surface width. Obtain expression or the critical depth and minimum specific energy for a wide rectangular.

3. The normal depth of flow of water in a rectangular channel 1.5 m wide is one meter. The bed slope of the channel is 0.0006 and Manning's roughness coefficient: 0.012. Find the critical depth. At a certain section of the same channel the depth is 0.92 while at a second section the depth is 0.86 m. Find the distance between the two sections. Also find whether the section is located d/s or u/s with respect of the first section.

OR

4. For a hydraulic jump in a rectangular channel, derive the following relationship. Also state the assumptions involved in the derivation. i) $y_1y_2(y_1+ y_2)/2 = q^2/g$.

5. State and discuss the assumptions made in the derivation of the dynamic equation for gradually varied flow. Starting from first principle, derive equation for the slope of the water surface in GVF. Also write down the dynamic equation for wide rectangular channels if (i) Manning's formula is used, and (ii) Chezy's formula is used.

OR

6. Sketch the possible GVF profiles in the serial arrangement of channels,if the flow is from left to right : (i) free intake –steep –sluice gate—mild slope, (ii) mild—sluice gate –steep—horizontal—sudden drop..

$H = z + y + v^2/2g$

Differentiating (H)

$dH/dx = dz/dx + dy/dx + d(v^2/2g)/dx$

we know $dz/dx = -S_0$

$dH/dx = -S_f$

$-S_0 - S_f = S_0 + dy/dx + dv^2/2g/dx$

$-dy/dx = S_0 - S_f - d(v^2/2g)/dx$

we know $v = Q/A$

$A = by$

So $v = Q/by = q/y$

$v^2/2g = (q/y^2)/2g = (q^2/2g \times y^2)$

Differentiate

$(v^2/2g)$

$d(v^2/2g)/dx = d/dx\,[q^2/2g \times y^{-2}]$

$= (q^2/2g) \times d(dy^{-2})/dx$

$= (q^2/2g) \times -2y^{-3} \times dy/dx$

$d(v^2/2g)/dx = q^2/(g \times y^3) \times dy/dx$

Substituting

$dy/dx = S_0 - S_f - (q^2/(g \times y^3)) \times dy/dx$

$dy/dx = S_0 - S_f + q^2/(g \times y^3)) \times dy/dx$

using manning formula

$V = 1/n \times R^{(2/3)} \times S_0^{(1/2)}$

$R \approx y \times 1.0$, $V = 1/n \times y^{2/3} \times S_0^{1/2}$

y $v = q/y$

$S_f = (n^2 \times q)^2 / y \times 10/3$

$dy/dx = \dfrac{S_0 - n^2 \times q^2 / y \times 10/3}{(1 - q^2/g \times y^3))}$

Chezy's Formula :

$v = c \times \sqrt{(R \times S_f}$

$R \approx y \cdot 1.0$, $v \approx c \times \sqrt{(y \times S}$

$v = q/y$

$q/y = c \times \sqrt{y \times S_f}$

$q/y = c^2 \times y \times S_f$

$S_f = q^2 \times /(c^2 \times y^3)$

$dy/dx = S_0 - q^2/c^2 \times$

$(1 - q^2/g \times y^3))$

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Rajiv Gandhi University of Knowledge Technologies- RK Valley-AP

Academic Year 2024 – 25: Semester 2 - Mid Semester Test-3

CIVIL & MECHANICAL - Introduction to Probability & Statistics

R21 Batch& Regular Mode of Examinations

Year:ENGG-2 | **Date of Examination: 11-04-2025**

Time:60 Minutes | **Max Marks:15**

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1. A multiple choice test consists of 8 questions with 3 answers to each question (of which only one is correct). A student answers each question by rolling a balanced die and checking the first answer if he gets 1 or 2 , the second answer if he gets 3 or 4 and the third answer if he gets 5 or6 . To get a distinction, the student must secure at least 75% correct answers. If there is no negative marking, what is the probability that the student secures a distinction?

(OR)

✓ 2. A manufacturer, who produces medicine bottles, finds that 0.1% of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the product of bottles. Find how many boxes will contain (i) no defective, and (ii) at least two defectives.

✓ 3. There are 600 engineering students in the B.Tech. classes of a university and the probability for any student to need a copy of a particular book from the university library on any day is 0.05 . How many copies of the book should be kept in the university library so that the probability may be greater than 0.90 that none of the students needing a copy from the library has to come back disappointed? Also find normal approximation to the binomial distribution.

(OR)

4. (a)

X is a normal variate with mean 30 and standard deviation 5. Find the probabilities that

(i) $26 \leq X \leq 40$ (ii) $X \geq 45$ (iii) $|X - 30| > 5$

(b) In a distribution exactly normal, 10.03% of the items are under 25kg weight and 89.97% of the items are under 70kg weight. What are the mean and standard deviatioin of the distribution?

5. Assume the sample variances to be continuous measurements. Find the probability that a random sample of 25 observations, from a normal population with variance $\sigma 2 = 6$, will have a sample variance S2

(a) greater than 9.1;

(b) between 3.462 and 10.745.

(OR)

6. Pull-strenght test in 10 soldered for a semiconductor device yield the following results, in pounds of force required to rupture the bond:

19.8	12.7	13.2	16.9	10.6
18.8	11.1	14.3	17.0	12.5

Another set of 8 leads was tested after encapsulation to determine whether the pull strength had been increased by encapsulation of the device, with the following results:

24.9	22.8	23.6	22.1	20.4	21.6	21.8	22.5

Comment on the evidence available concerning equality of the two population variances.

Rajiv Gandhi University of Knowledge Technologies- RK Valley -AP

Academic Year 2024-25: Semester II - Mid Semester Test-III

Civil Engineering - 23CE2204 – Structural Analysis

R21 Batch& Regular Mode of Examinations

Year: ENGG-2 | **Date of Examination: 10 APRIL, 2025**

Time: 60 Minutes | **Max Marks: 15**

3X5 Marks= 15

The Question Paper Consists of 2 units. Answer one full question from each unit. Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

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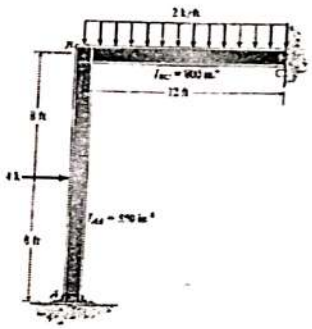
1. Using moment distribution method, determine the moments at each end of each member. Draw the moment diagram. Let E = 29000 ksi. The moment of inertia of each member is shown on the figure above. Assume that at joint B is rigid, C is pinned and A is fixed.



OR

2. A continuous beam ABC fixed at the ends is loaded as shown in figure. Find the reactions and support moments. Draw also B.M and S.F diagrams.

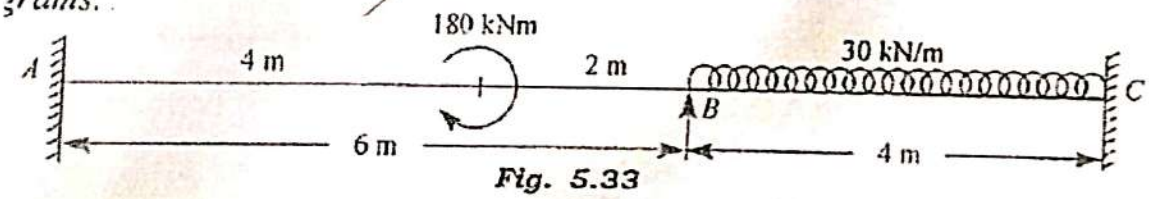


Fig. 5.33

3. Two wheel loads 200kN and 80kN spaced at 2m apart move on the span of girder of span 16 meters. Find the maximum bending moment that occur at section 6 meters from the left end. Any wheel load can lead the other.

OR

4. The load system shown in Fig. moves from right to left on a girder of span 20 meters. Find the maximum shear force at a section 7.5 meters from the left end.

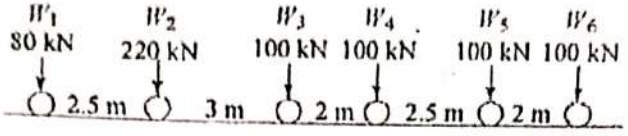
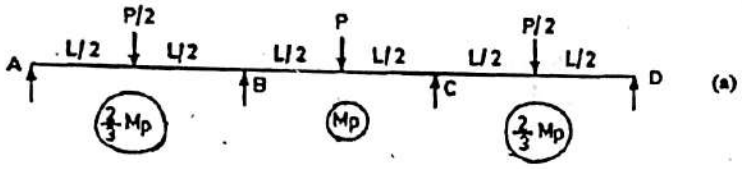


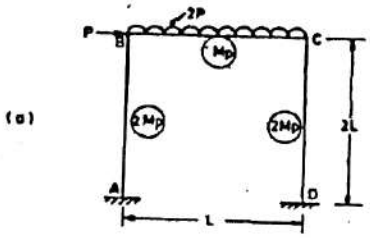
Fig. 11.40

5. Find the collapse load for the three span continuous beam as shown in fig. The plastic moment of the end span sections is two third of the plastic moment capacity (M_p) of the mid span section.



OR

6. Find the collapse load and draw a bending moment diagram at collapse for the portal frame is shown in figure.



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Section-B (100 Minutes)

This section contains 5 questions. Solve any four questions out of five. Each Question carries 10 marks. Total marks to this section are 40.

1. An orifice meter consisting of 100 mm diameter orifice in a 250 mm diameter pipe has $C_d = 0.65$. The pipe delivers oil of specific gravity 0.8. The pressure difference in the two sides of the orifice plate is measured by mercury-oil differential manometer. If the differential gauge reading is 800 mm of mercury, find the rate of flow of oil through the pipe in litres/s.

2. A pipe carrying water has a 30 cm x 15 cm venturimeter which is positioned inclined at 30° to the horizontal. The flow is upwards. The converging cone is 45 cm in length and the coefficient of discharge of the meter is 0.98. A differential U tube manometer with mercury as indicating fluid is connected to the inlet and to the throat and shows a differential column height of 30cm.
   i. Calculate the discharge in the pipe
   ii. If the pressure at the inlet section is 50 kPa, determine the pressure at the throat.
   iii. Find the head loss in the converging section of the venturi meter.

3. Do the sample calculation of V-notch for given data suitable for our lab experimental set up.

   Height of flow in notch above crest or sill level: 0.3 m

   Angle of notch: 60^0

   Time took to fill 10cm of height of fluid in the tank: 20 s.

   Tank measurements: length = 30cm, breadth = 30cm.

   Find the discharge coefficient?

4. Write Expression for Bernoulis principle. Write any three applications of Bernoullis Theorem? Explain the working and principle of Pitot tube?

5. A discharge of 10 lit/sec is flowing through a pipe of diameter 25mm and of length 1m. The head loss between the two ends of pipe is 4m of water. Calculate the value of coefficient of friction for the given pipe.

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Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester II - Mid Semester Test- III
CIVIL Engineering - 23CE2202-Environmental Engineering-I
R21 Batch & Regular Mode of Examinations

Year: ENGG-2 | **Date of Examination: 9th April,2025.**
Time: 60 Minutes | **Max Marks: 15**

3X5 Marks= 15

The Question Paper consists of 3 units. Answer one full question from each unit.
Each Question Carries 5 Marks and may have 'A', 'B' as two sub-questions

1) Explain the differences between lime soda process and zeolite process?

(OR)

2) Explain demineralization process and it's advantages and disadvantages?

3) Explain the treatments used for removal of
   a)Taste, colour and odour (3M)
   b) Removal of iron and manganese (2M)

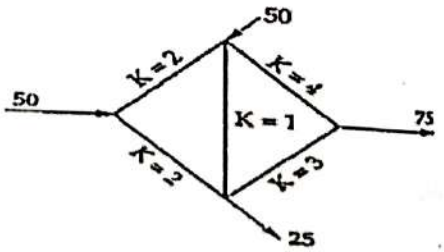
(OR)

4) Derive the expression for EQUIVALENT LENGTH OF PIPE when the pipes are arranged in series and parallel by EQUIVALENT PIPE METHOD?

5) Describe in detail various types of distribution layouts with it's advantages and disadvantages?

(OR)

6) Determine the distribution of flow in the pipe network shown in figure. The head loss H_L may be assumed as KQ^n. The flow is turbulent and pipes are rough. The value of K for each pipe is indicated in the figure. Use hardy-cross method?



Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester 2 - Mid Semester Test-3
Civil Engineering - 20CE 2203 – Geotechnical Engineering I
R21 Batch & Regular Mode

R210517

Year: ENGG-2
Time: 60 minutes

Date of Examination: 10-April- 2025
Max Marks: 15

3X5 Marks= 15

The Question Paper consists of 6 questions with internal choice. Answer any 3 questions from each set. Each Question Carries 5 Marks

Note: Usage of Scientific calculator may be permitted

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1. Define quick sand condition. Derive an equation for the hydraulic gradient in the quick sand condition with a neat sketch.

OR

2. A sand deposit is 10 m thick and overlies a bed of soft clay. Water table is at 3 m depth from the ground surface. If sand above the water table has a degree of saturation of 45%, plot the diagram showing variation of total stress, neutral stress and effective stress. e of sand is 0.7 and Gs = 2.65. Also calculate the change in effective stress if water table is lowered by 5 m.

3. A soil profile consists of a surface layer of clay 4 m thick having unit weight of 19.5 kN/m^3 and a sand layer of 2 m thick having unit weight of 1.85 g/cc overlying an impermeable rock. Water table is at the ground surface. If water table in a stand pipe driven into the sand layer rises 2 m above the ground surface, draw the plot showing variation of σ, u, and σ'. Also determine the increase in σ' at the top of rock when the artesian head in sand is reduced by 1 m.

OR

4. Explain the procedure to determine field density of soil by core cutter method. Also define relative compaction and write its formula.

5. Explain initial, primary and secondary consolidation in detail.

OR

6. Explain the dial guage reading and time plot curve in the case of sand and clay. Also explain the final void ratio and effective stress plot.

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Rajiv Gandhi University of Knowledge Technologies - RK Valley- AP

Academic Year 2024 – 25: Semester II - MidTest-3

Civil Engineering - 23CE2205- Water Resources Engineering R210517

R21 Batch & Regular Mode of Examinations

Year: ENGG-2 **Date of Examination: 11th April, 2025**

Time: 60 Minutes **Max Marks: 15**

The Question Paper consists of 3 sections. Answer three question from the given six questions. **3X5 Marks= 15**

Each Question Carries 5 Marks

Note: Usage of scientific calculator is permitted

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1. A 200 mm diameter well fully penetrates in a confined aquifer 30 m depth for a steady rate of pumping of 35 lit/sec. The steady drawn at 100 m and 300 m distance from the well are observed as 3.2 m and 2.5 m respectively. Estimate the hydraulic conductivity (K) in m/day and transmissibility (T) of the aquifer in m^2/day and also estimate the draw down at the well.

OR

2. Route the following flood hydrograph through a river reach for which Muskingum coefficient K= 8 h and x = 0.25.

Time (hr)	0	4	8	12	16	20
Inflow (m^3/s)	8	16	30	25	15	10

The initial outflow discharge from the reach is 8.0 m^3/s.

3. A 30 cm well completely penetrates an unconfined aquifer of saturated depth 40 m. After long period of pumping at a steady rate of 1500 lpm, the drawdown in two observation wells at 25m and 75m from the pumping well were found to be 3.5m and 2.0m respectively. Determine the transmissivity of the aquifer. What is the drawdown at the pumping well?

OR

4. A confined horizontal aquifer of 15m thickness and permeability K= 20 m/day connects two reservoirs M and N situated 1.5 km apart. The elevation of the water surface in reservoirs M and N measured from datum or impervious layer is 45 m and 25 m respectively. If the reservoir M is polluted by a contaminant suddenly, how long will it take the contaminant to reach the reservoir N? Assume the porosity of the aquifer n=0.30.

5. Describe the numerical methods of hydrologic reservoir routing?

OR

6. Explain in briefly the following terms used in groundwater studies:
a). Specific yield, b) Storage coefficient, c) Specific Capacity